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(54) **DUST-PROOF FILTER STRUCTURE FOR POWER SUPPLY PANEL**

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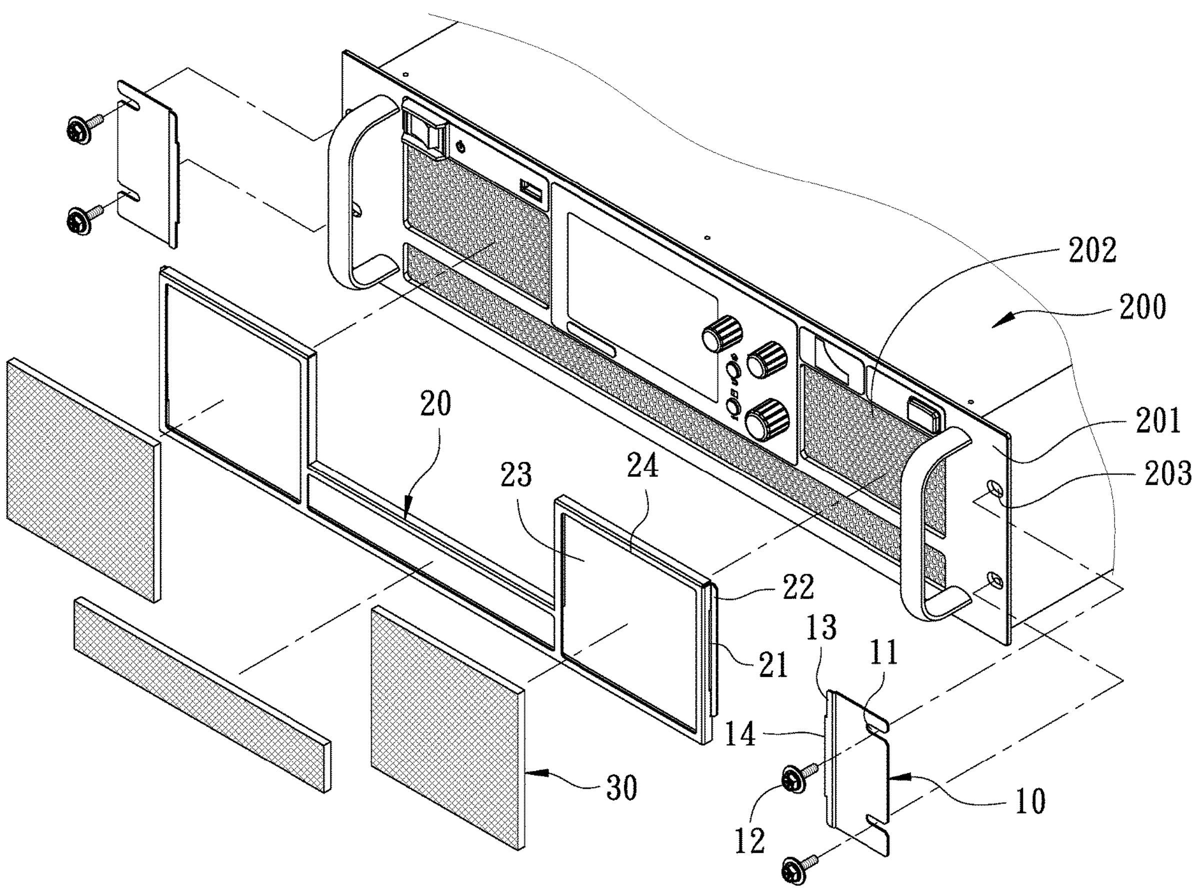
(57) **ABSTRACT**

A dust-proof filter structure for a power supply panel is disclosed. The dust-proof filter structure is mounted to a front of a panel of a power supply. The panel has at least one air inlet hole. The dust-proof filter structure includes two retaining plates locked on two sides of the front of the panel. A frame is connected between the retaining plates and located on the front of the panel. The frame has a window corresponding in position to the air inlet hole. A detachable filter cotton mesh is disposed in the window. Thus, the cooling air flowing into the power supply can be filtered to prevent the components inside the power supply from being contaminated, so as to prolong the service life of the power supply and maintain the normal operation of the components. Besides, the filter cotton mesh can be quickly mounted and easily removed for cleaning.



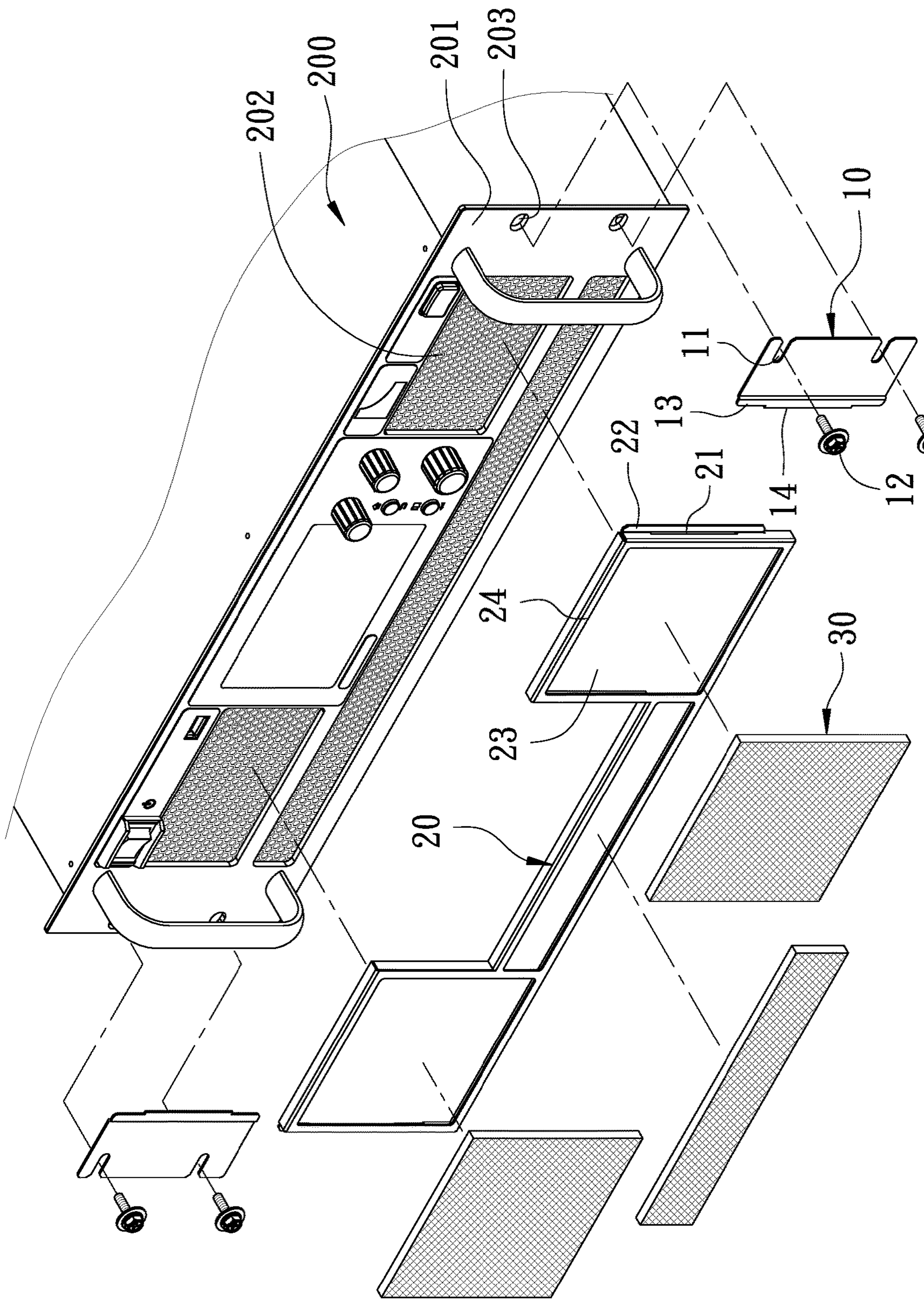


FIG. 1

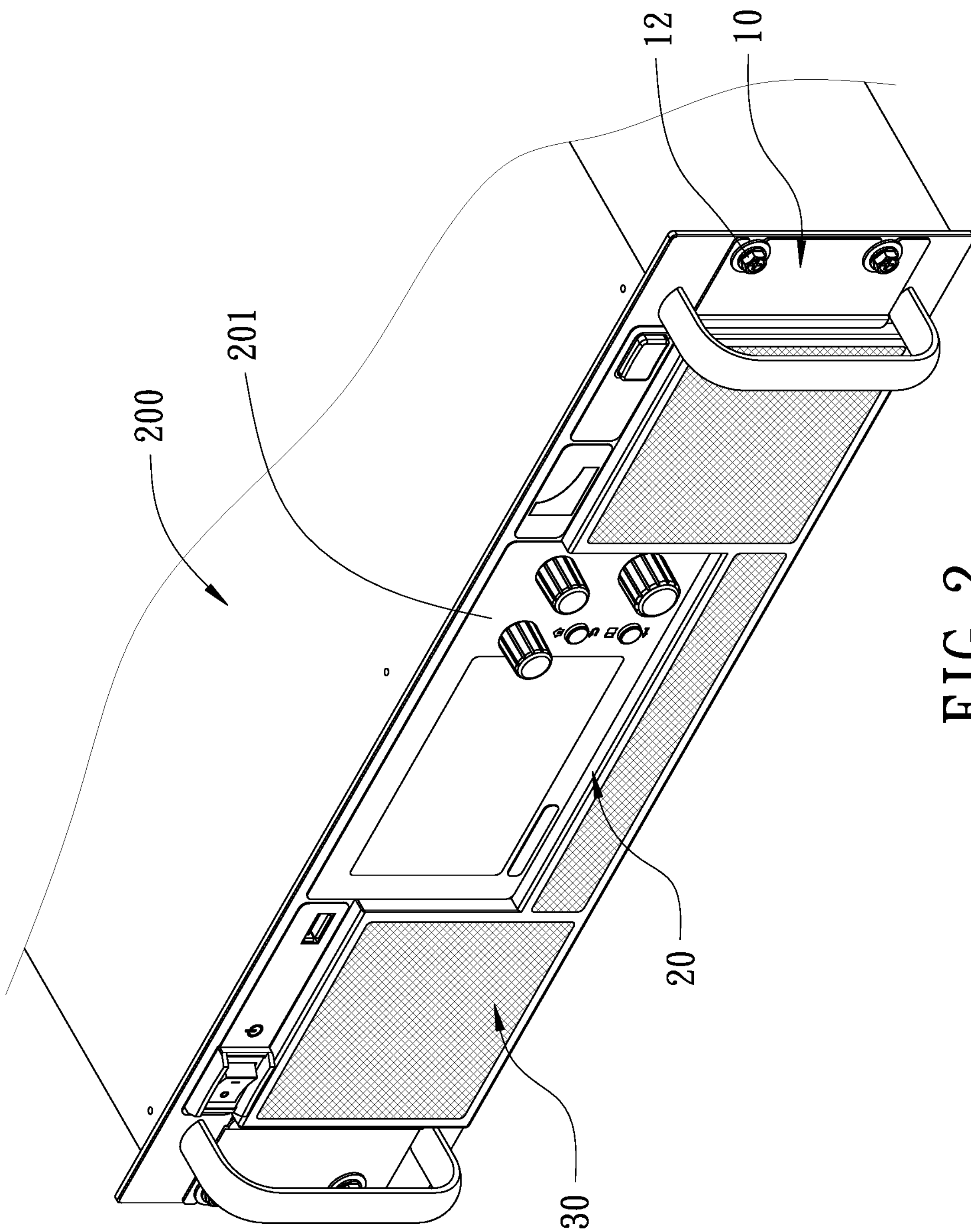


FIG. 2

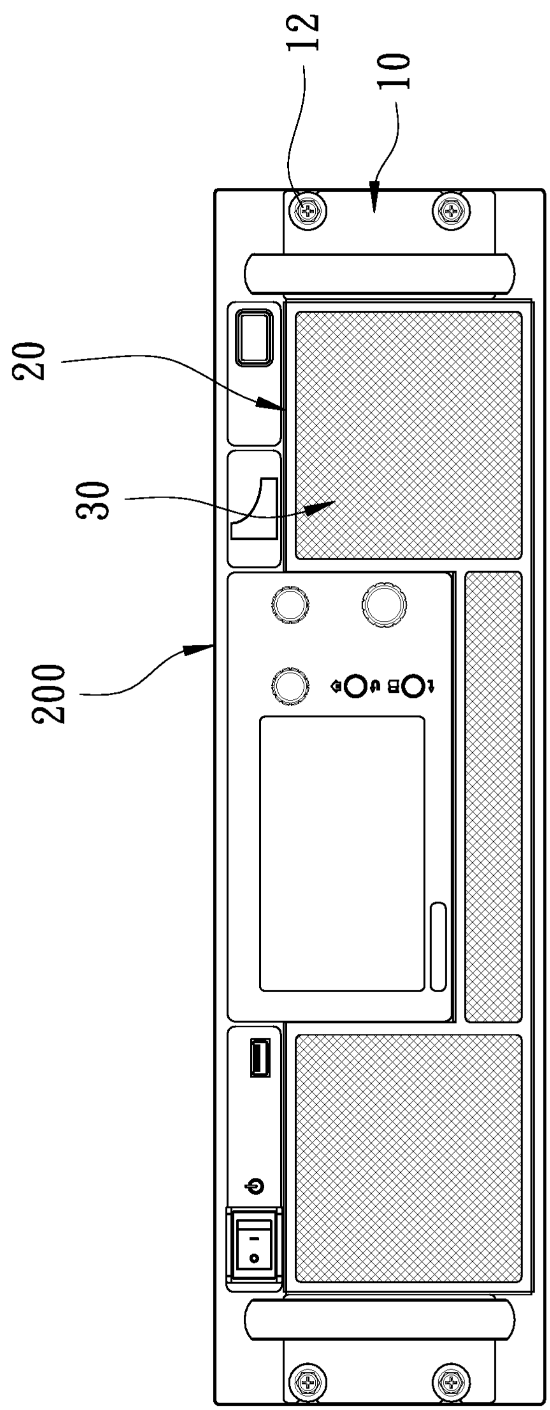


FIG. 3

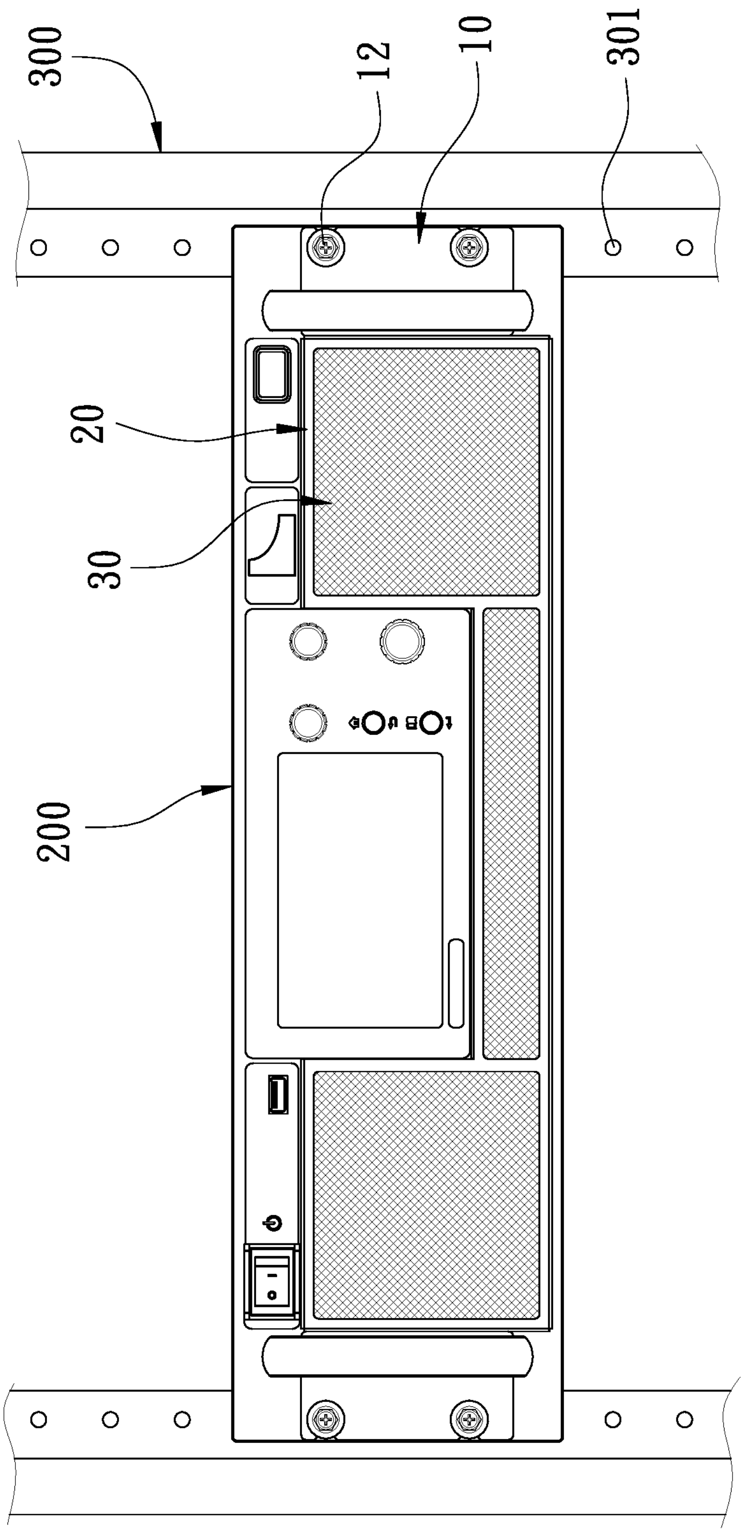


FIG. 4

DUST-PROOF FILTER STRUCTURE FOR POWER SUPPLY PANEL

FIELD OF THE INVENTION

[0001] The present invention relates to a power supply, and more particularly, to a dust-proof filter structure for a power supply panel.

BACKGROUND OF THE INVENTION

[0002] In general, most of machines and equipment that use air for cooling have an internal fan to draw the external air via air inlet holes for cooling the internal parts. However, due to the deterioration of air quality, the air contains impurities, lint, dust and chemical substances, and the impurities are easy to accumulate on the parts. When electronic components and other related parts are exposed to dirt for a long time, they may malfunction or being damaged easily.

[0003] Especially for a power supply, it utilizes a fan inside the power supply to draw the external air via air inlet holes for heat dissipation of the internal electronic components. It is well known that the power supply includes many electronic components, such as circuit boards, capacitors, coils and power transistors, etc. If the electronic components are covered with dirt for a long time, in addition to being unable to dissipate heat effectively, they may lose functionality or malfunction. The power supply generally uses an internal fan to directly draw the external air via air inlets for heat dissipation, without any means to prevent dust from entering the power supply. Especially in poor environments such as factories, the pollution will be more serious after a long period of time.

[0004] Accordingly, the inventor of the present invention has devoted himself based on his many years of practical experiences to solve these problems.

SUMMARY OF THE INVENTION

[0005] The primary object of the present invention is to provide a dust-proof filter structure for a power supply panel. Through the dust-proof filter structure, the cooling air flowing into the power supply can be filtered to prevent the components inside the power supply from being contaminated, so as to prolong the service life of the power supply and maintain the normal operation of the components. Besides, the dust-proof filter structure has a filter cotton mesh that can be quickly mounted and easily removed for cleaning.

[0006] In order to achieve the foregoing object, the present invention provides a dust-proof filter structure for a power supply panel. The dust-proof filter structure is mounted to a front of a panel of a power supply. The panel has at least one air inlet hole. The dust-proof filter structure comprises two retaining plates, a frame, and at least one filter cotton mesh. The two retaining plates are locked on left and right two sides of the front of the panel, respectively. The retaining plates each have an engaging piece on an inner side thereof. The frame is located on the front of the panel. Two sides of the frame have engaging holes corresponding to the engaging pieces of the retaining plates, respectively. The engaging pieces of the retaining plates are engaged in the respective engaging holes, so that the frame is combined with the retaining plates and secured to the front of the panel. The frame has at least one window corresponding in position to the air inlet hole. The window corresponds in shape to the air inlet hole. The filter cotton mesh corresponds in shape to the window and is disposed in the window.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** is an exploded view of a preferred embodiment of the present invention;

[0008] FIG. **2** is a perspective view of the preferred embodiment of the present invention;

[0009] FIG. **3** is a front view of the preferred embodiment of the present invention; and

[0010] FIG. **4** is a schematic view of the preferred embodiment of the present invention mounted to a rack.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0011] Embodiments of the present invention will now be described, by way of example only, with reference to the accompanying drawings.

[0012] FIG. **1** is an exploded view of a preferred embodiment of the present invention. FIG. **2** is a perspective view of the preferred embodiment of the present invention. FIG. **3** is a front view of the preferred embodiment of the present invention. The present invention discloses a dust-proof filter structure for a power supply panel. The dust-proof filter structure is mounted to the front of a panel **201** of a power supply **200**. The panel **201** has a plurality of air inlet holes **202** and a plurality of locking holes **203**. The locking holes **203** are located the left and right two sides of the panel **201** for the panel **201** to be locked to a rack.

[0013] The dust-proof filter structure includes two retaining plates **10**, a frame **20**, and a plurality of filter cotton meshes **30**.

[0014] The two retaining plates **10** are locked on the left and right two sides of the front of the panel **201**, respectively. The retaining plates **10** are formed with a plurality of horizontal elongate holes **11** corresponding to the respective locking holes **203**. The elongate holes **11** each have an open outer end, so that the retaining plates **10** can be mounted and dismounted easily. A plurality of bolts **12** lock the retaining plates **10** to the panel **201**. The inner side of each retaining plate **10** is bent to form a vertical shoulder portion **13**. The inner side of the shoulder portion **13** has an engaging piece **14**.

[0015] The frame **20** is located on the front of the panel **201**. Two sides of the bottom of the frame **20** extend outwardly to from vertical extension plates **22** corresponding to the shoulder portions **13** of the retaining plates **10**, respectively. The extension plates **22** are inserted into and restricted by the shoulder portions **13** of the retaining plates **10**, respectively. Two sides of the frame **20** have engaging holes **21** corresponding to the engaging pieces **14** of the retaining plates **10**, respectively. The engaging pieces **14** of the retaining plates **10** are engaged in the respective engaging holes **21**, so that the frame **20** is retained by the retaining plates **10** and secured to the front of the panel **201**. The frame **20** has windows **23** corresponding in position to the air inlet holes **202**. The windows **23** correspond in shape to the respective air inlet holes **202**. The surface of the frame **20** has a stop edge **24** protruding from the periphery of the windows **23**.

[0016] The filter cotton meshes **30** each have a flexible mesh body for filtering out impurities, lint and dust in the air. The filter cotton meshes **30** correspond in shape to the respective windows **23** and are inserted and placed in the windows **23**. The filter cotton meshes **30** are confined by the stop edge **24**, so as to be secured in the windows **23**.

[0017] In order to further understand the structural features, the technical means and the expected effects of the present invention, the assembly and effects of the present invention are described hereinafter.

[0018] When the dust-proof filter structure provided by the present invention is to be mounted on the panel **201** of the power supply **200**, first, the engaging pieces **14** of the retaining plates **10** are engaged in the respective engaging holes **21** on the two sides of the frame **20**, so that the frame **20** is combined with the retaining plates **10**. Then, the bolts **12** pass through the elongate holes **11** of the retaining plates **10** and are locked to the locking holes **203** of the panel **201**. Thus, the frame **20** and the retaining plates **10** are secured to the front of the panel **201**. Finally, the filter cotton meshes **30** are placed in the corresponding windows **23** respectively. Because the filter cotton meshes **30** are flexible, they can be slightly bent to be placed in the windows **23** and confined by the stop edges **24** to achieve a positioning effect, thereby completing the assembly of the dust-proof filter structure.

[0019] FIG. **4** is a schematic view of the preferred embodiment of the present invention mounted to a rack. When the power supply **200** of the present invention is to be installed on a rack **300**, the bolts **12** are inserted through the elongate holes **11** of the retaining plates **10** and the locking holes **203** of the panel **201** and locked to positioning holes **301** of the rack **300**.

[0020] From the above description, through the dust-proof filter structure provided by the present invention, the cooling air flowing into the power supply **200** can be filtered to prevent the components inside the power supply **200** from being contaminated, so as to prolong the service life of the power supply **200** and maintain the normal operation of the components. Besides, the filter cotton meshes **30** can be quickly mounted and easily removed for cleaning. In addition, since the anti-dust filter structure provided by the present invention is installed and secured via the locking holes **203** that are originally used for being secured to the rack **300**, the panel of the power supply **200** does not need additional molding or locking holes, thereby saving the cost of materials.

[0021] Although particular embodiments of the present invention have been described in detail for purposes of illustration, various modifications and enhancements may be made without departing from the spirit and scope of the present invention. Accordingly, the present invention is not to be limited except as by the appended claims.

What is claimed is:

1. A dust-proof filter structure for a power supply panel, mounted to a front of a panel of a power supply, the panel having at least one air inlet hole, the dust-proof filter structure comprising:

two retaining plates, locked on left and right two sides of the front of the panel respectively, wherein the retaining plates each have an engaging piece on an inner side thereof;

a frame, located on the front of the panel, two sides of the frame having engaging holes corresponding to the engaging pieces of the respective retaining plates, the engaging pieces of the retaining plates being engaged in the respective engaging holes so that the frame is combined with the retaining plates and secured to the front of the panel, wherein the frame has at least one window corresponding in position to the air inlet hole, and the window corresponds in shape to the air inlet hole;

at least one filter cotton mesh, corresponding in shape to the window and being disposed in the window.

2. The dust-proof filter structure as claimed in claim **1**, wherein two sides of the panel have a plurality of locking holes, a plurality of bolts pass through the locking holes for locking the power supply to a rack, the retaining plates are formed with a plurality of horizontal elongate holes corresponding to the respective locking holes for the bolts to lock the retaining plates to the panel.

3. The dust-proof filter structure as claimed in claim **2**, wherein the elongate holes each have an open outer end.

4. The dust-proof filter structure as claimed in claim **1**, wherein a surface of the frame has a stop edge protruding from a periphery of the window, so that the filter cotton mesh is confined by the stop edge to be secured in the window.

5. The dust-proof filter structure as claimed in claim **1**, wherein the inner side of the retaining plate is bent to form a vertical shoulder portion, two sides of a bottom of the frame extend outwardly to from vertical extension plates corresponding to the shoulder portions of the retaining plates, and the extension plates are inserted into and restricted by the shoulder portions of the retaining plates, respectively.

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